

Customer Churn Predictor Process Report

Project Overview

The goal of this project was to develop and deploy an interactive Streamlit web application for predicting customer churn risk, hosted on Hugging Face Spaces at <https://huggingface.co/spaces/Mr0Diablo/customer-churn-predictor>. The app allows users to upload a CSV file containing customer data, validates the data, performs churn predictions using a pre-trained machine learning model, and presents results through visualizations and downloadable outputs.

Objectives

1. Create a user-friendly interface for uploading and analyzing customer data.
 2. Integrate a pre-trained Logistic Regression model for churn prediction.
 3. Implement data validation to ensure compatibility with the model's requirements.
 4. Provide insightful visualizations using Plotly for prediction results.
 5. Deploy the app on Hugging Face Spaces for public access.
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Development Process

Environment Setup

- Tools and Libraries: Installed Python 3.8+, Streamlit, Pandas, NumPy, Pickle, Plotly, and other dependencies.

- **Model Preparation:** Obtained the pre-trained model (`complete_churn_pipeline.pkl`), which includes a Logistic Regression classifier and preprocessing pipeline.
- **Development Environment:** Used a local development environment with VS Code and a virtual environment to manage dependencies.

App Structure and Design

- **Streamlit Configuration:** Set up the app with a wide layout, custom page title, and icon using `st.set_page_config`. Implemented a dark theme via custom CSS for better aesthetics.
- **Sidebar:** Designed a sidebar for file uploads, model information, usage tips, and developer credits, enhancing navigation and user experience.
- **Main Interface:** Divided the interface into sections for data upload, data overview, and churn prediction, using tabs for organized content display.
- **CSS Styling:** Created custom styles for headers, buttons, cards, and metric displays to ensure a cohesive dark-mode appearance.

Core Functionality

- **Model Loading:** Implemented a cached function (`load_model`) to load the pickled model efficiently, with error handling for missing files.
- **Data Validation:** Developed a `verify_data` function to check for required features (e.g., age, gender, points_in_wallet) in the uploaded CSV.
- **Prediction Logic:** Created a `predict_churn` function to generate predictions and probabilities using the model, with error handling for invalid inputs.
- **Results Display:** Built a `display_predictions` function to present results in two tabs:
 - **Overview:** Shows summary metrics (total customers, churn risk distribution) and a Plotly pie chart.
 - **Detailed Results:** Displays a DataFrame with predictions and probabilities.

- Data Visualizations: Added interactive Plotly charts (histogram for numeric features, bar chart for categorical features) to explore uploaded data.
- Download Option: Enabled users to download prediction results as a CSV file.

Challenges and Solutions

- Challenge: Ensuring the pie chart displayed both churn risk categories (Low Risk, High Risk) even if one was absent in predictions.
 - Solution: Explicitly included both categories in the pie chart data structure with zero counts for missing categories, using a color map for consistency.
- Challenge: Handling missing or incorrect columns in uploaded CSVs.
 - Solution: Implemented robust data validation with clear error messages and a warning box to guide users.
- Challenge: Achieving a responsive and visually appealing dark theme.
 - Solution: Used custom CSS to override Streamlit's default styles, ensuring consistent colors and layouts across components.
- Challenge: Optimizing model loading for performance.
 - Solution: Utilized Streamlit's `@st.cache_resource` to cache the model, reducing load times.

Testing

- Local Testing: Ran the app locally (`streamlit run app.py`) to test functionality, UI, and error handling.
- Data Testing: Used sample CSVs with correct, missing, and incorrect columns to verify validation and prediction logic.
- Visualization Testing: Ensured Plotly charts rendered correctly across different data distributions.

- Edge Cases: Tested scenarios like empty files, large datasets, and missing model files to ensure robust error handling.
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Deployment on Hugging Face Spaces

1. Preparation

- Repository Setup: Created a Hugging Face Space repository (Mr0Diablo/customer-churn-predictor).
- File Structure:
 - streamlit_app.py: Main Streamlit app script.
 - complete_churn_pipeline.pkl: Pre-trained model file.
 - requirements.txt: Listed dependencies (e.g., streamlit, pandas, plotly, scikit-learn).
- Dependence: Ensured requirements.txt included exact versions compatible with Hugging Face's Python environment.

2. Deployment Steps

3. Pushed the app files to the Hugging Face Space repository via Git.
4. Configured the Space to use the Streamlit framework in the Hugging Face interface.
5. Monitored the build process in the Space's logs to resolve any dependency or runtime issues.
6. Tested the deployed app at <https://huggingface.co/spaces/Mr0Diablo/customer-churn-predictor> to verify functionality.

7. Post-Deployment

- Validation: Confirmed that the app loaded the model, processed uploads, and displayed results correctly.
- Performance: Checked responsiveness and loading times on the Hugging Face infrastructure.
- Updates: Planned for future updates to add features like model retraining or additional visualizations.

Outcomes

- Successfully deployed a functional Streamlit app on Hugging Face Spaces.
- Provided an intuitive tool for businesses to predict customer churn and explore data.
- Achieved a visually appealing, dark-themed interface with interactive visualizations.
- Ensured robust error handling and data validation for user-friendliness.

Future Improvements

- Add feature importance visualizations to explain model predictions.
- Implement batch processing for larger datasets.
- Integrate real-time data streaming for dynamic predictions.
- Enhance accessibility with light theme options and keyboard navigation.

Conclusion The development and deployment of the Customer Churn Predictor app demonstrated the effective use of Streamlit for building interactive machine learning applications. By addressing technical challenges and focusing on user experience, the project delivered a valuable tool for customer retention analysis, accessible to a wide audience via Hugging Face Spaces.

Developed By: Churn Project Team Last Updated: May 2025